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ICE 9A.1



Aman Gupta · Apr 10, 2025 (Edited Apr 11, 2025)

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1. (1 mark) Let $X = [x_1, x_2, x_3, x_4]$, where x_i is an 8 bit character with its 8th bit set to even parity. Now consider the hash function, $H(X) = [p_1, p_2, p_3, p_4]$, where p_i is the parity bit corresponding to x_i . Does $H(X)$ satisfy the property called "Pre-image resistance"? In other words, given a 4 bit string (call it t) is it computationally impossible to find X such $H(X) = t$?

- YES
- **NO (Correct)**
- Cannot tell

2. (1 mark) Revisit the problem given in 1 above. Does the hash function $H(X)$ satisfy the property called "Second Pre-image resistance"? In other words, given X , is it computationally impossible to find Y such $H(X) = H(Y)$?

- Yes
- **No (Correct)**
- Cannot Tell

3. (1 mark) Revisit the problem given in 1 above. Does $H(X)$ satisfy the property called "Strong resistant property"? In other words, is it computationally impossible to find a pair of X and Y such that $H(X) = H(Y)$?

- Yes
- **No (Correct)**
- Cannot tell

4. (1 mark) Refer to the scheme below that tries to ensure both message authentication and integrity. What would happen if the hashing function $h(.)$ does not satisfy the FIRST PRE-IMAGE resistance property? That is, which of the statements below best describes the situation?

- **The secret S may be decoded (Correct)**
- The message M that was sent may be substituted by an intruder with another message M_2
- A user may construct two messages M_1 and M_2 that hash into the same hash value. Further, he may send M_1 , but claim that he actually sent M_2
- None of the above

5. (2 marks) Refer to the scheme below that tries to ensure both message authentication and integrity. What would happen if the hashing function $h(.)$ does not satisfy the SECOND PRE-IM/



resistance property? That is, which of the statements below best describes the situation?

- Confidentiality of message communication may be compromised
- **Message integrity may be compromised during communication (Correct)**

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6. (2 marks) Refer to the encryption/hashing scheme given in question 5 again that tries to ensure both message authentication and integrity. Assume that the hashing function $h(.)$ is a cryptographically strong hashing function. Does the scheme ensure Non-Repudiation service? That is, which of the statements below best describes the situation?

- Non-repudiation service is not supported by this scheme, since the hashing function may or may not be strongly collision resistant
- **Non-repudiation service is not supported by this scheme because of the use of symmetric-key encryption/decryption (Correct)**
- Non-repudiation service is indeed supported
- None of the above

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